Multi-Agent Research Factory: Design Document

1. Architecture & Agent Specialization

The Planner (Decomposition), Researcher (Search), Analyst (Synthesis), Critic (Validation), Writer (Output), and Coordinator (State management).

2. Inter-Agent Communication & Replanning Loop

Event-driven pub/sub message bus. Replanning triggered by Critic rejection (Accuracy < 85%, Rigor < 25%, Consensus 20%).

3. Data Flow & Security Guarantees

Immutable execution traces, schema-validated payloads (Zod/JSON Schema), local sandbox execution.